Reframing the Role of Novelty within Social HRI: from Noise to Information

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Abstract **– Social robots, understood as the category of embodied robots extending into social domains through reciprocal social interaction, are still a practical novelty in most of these domains today. However, the phenomenon of novelty effects is only eclectically and peripherally addressed within most research into social human-robot interaction, and even when treated more extensively, it is usually framed as a source of noise in need of reduction. In this paper, I will argue a reframing of novelty effects that posits the phenomenon as a valuable source of information. In the first part of the paper, I present a tentative account of what I call experiential novelty in order to illustrate (1) that novelty should be understood as an 'original feature of experience', (2) that novelty arises in the engagement between an experiencer and an experience where the experiencer's possessed knowledge is inadequate in making sense of the experience, and (3) that novelty effects should be seen as cognitive and behavioural expressions of a 'search for meaning'. In the latter part of the paper, I discuss some of the current research lines within social human-robot interaction research from the perspective of this account of novelty. Most notably, I argue that retrospectively, the account holds explanatory utility in analyzing many of the findings in this research-field, and prospectively, the account holds generative utility in pointing to new ways in which participant experiences of novelty may be employed in research.**

Keywords— novelty effects; experiential novelty; social human-robot interaction; social robots; engagement; sense-making; expectations; learning; familiarization;

I. Introduction

> Thus a new technological genre may be emerging that challenges traditional ontological categories (e.g., between animate and inanimate). If we are correct, then it may be that the English language is not yet well equipped to characterize or talk about this genre. […]. Similarly, it may not be the best approach to keep asking if this emerging technological genre is, for example, "alive" or "not alive" if from the person's experience of the subject-object interaction, the object is alive in some respects and not alive in other respects, and is experienced not simply as a combination of such qualities […] but as a novel entity. [1]

Fourteen years have passed since Kahn and colleagues presented this reflection. Since then, research into social human-robot interaction (SHRI) has increased rapidly, both in the number of researchers, disciplines, methodologies, and funds invested in the enterprise, as well as in the number of conferences, publications, and studies generated. Though differing in aims, perspectives, and application areas, the efforts subsumed under the umbrella of SHRI share the overarching enterprise of investigating the promises and limitations of this novel agent-category entering the social scene. On basis of this alone, it does not seem unreasonable to have the impression that our efforts as researchers in this area are increasingly being guided to a larger extent by well-researched hypotheses and accumulated knowledge than by mere explorational 'trial-and-error' based on imaginaries. However, the fact still stands that social robots have *not*, within these last 14 years, become a familiar or commonly integrated part of the social landscapes within most societies today (although, arguably more so within some societies than others [2]) [3]. Though the concept of a social robot has been widely explored within popular fiction, most individuals have never actually interacted with such a social, robotic agent; and even when they have, it has most likely not been for very long. Thus, irrespective of whether or not social robots may be an *ontologically* new category (which I will discuss in some detail later), this category of social agents does still pose a novelty to most people today, at least from the basic circumstance of a lack of interactive experience with them.

Following the emergence of SHRI as an established research field, an emphasis on *process* and *ecology* has equally become more salient. Several have voiced the need for moving out of the laboratories and into the 'real world' [4]; for focusing on the processes influencing the investigated interactions *in medias res* [5, 6, 7]; and for developing studies that integrate various disciplines, perspectives, and/or frameworks [4, 8, 9], in order to produce data that are better suited for guiding the longer-term development and integration of social robots. Long-term studies [5], investigations of experiential dimensions [7], ethnographic studies[10, 11], multidisciplinary designs [9], user-driven design approaches [4] and meta-reflections of the generative role of the field itself [12, 13], are all examples of this emphasis.

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However, despite the fact that these efforts certainly enhance the visibility of potential novelty effects at large, such effects have only been eclectically, if not rarely, addressed at more detailed levels than what commonsensical notions have to offer.

To offer representative examples, in [14] the researchers hypothesize that their finding of an observed decrease of interaction over time was probably due to the novelty of the robot initially causing unreasonable expectations that lead to disappointment. In [8], where Dautenhahn investigates the dimensions of HRI research and proposes a conceptual space for this research, she briefly notes that "The attitude towards and opinion of such a machine will be biased by 'first impressions', but will change during long-term experiences". She further explores this in a collaborative paper [15] where the aim is to explicitly investigate habituation effects on participant preferences. Here, the authors conclude the paper by stating that "Therefore in this paper, it is strongly argued that it is essential to conduct long-term studies, to identify the social behavior parameters that are influenced by habituation effects in order to create better social behavior models for robots that adapt their social rules over time". Interestingly, in another paper published around the same time by most of the same authors, habituation or novelty effects are not mentioned [16]. In 2013, [5] present a 'survey on the existing long-term studies involving social robots', where the phenomenon of novelty effects consistently receives attention as a main reason for conducting long-term studies. The authors propose a model for how to address potential novelty effects, for instance stating that:

> Concerning the number of users interacting with the robot at the same time, we argue that if the robot interacts with a group of users, users need more time for the novelty effect to fade away, since the robot will be switching its attention by the different users. As for the complexity of the robot's behaviour, if the repertoire of the robot is more limited, it is more likely that the novelty effect will fade away more quickly. [5]

In another study from 2017, effects of novelty also receive consistent attention, this time in relation to why some people refuse to use or abandon use of robots [17]. While the researchers offer many valuable insights on the importance of investigating the reasoning behind choosing to either continue or abandon use of a robotic technology *after* initial use trials, their notion of novelty remains superficial. In identifying different motivations for non-use, the category 'End of novelty' is characterized simply in terms of heightened exploration that will decrease as novelty fades. As many of the other identified motivations seem to be effects of familiarization, this also suggests that the researchers do not operate with a close relation between novelty and familiarization.

Several studies have also focused on the role of prior knowledge of participants, but most operate with a relatively superficial notion of *how* (a lack of) prior knowledge may affect participant-responses; an example is [18] where the researchers investigate the relation between unfamiliarity and participant experiences of discomfort. However, their finding that there was no significant relation could be explained by a failure to acknowledge the practical dimensions of novelty experiences (which I will elaborate in sections 2 and 3) – an acknowledgment that [19] indirectly touch upon, when proposing that their failure to observe an attitude-change in their participants over time may be due to the fact that the experiment design did not require the participants to make practical sense of the robot.

Although far from exhaustive, what can be gleaned from these examples is that although novelty effects may be (directly or indirectly) addressed, these addressings seem to do little to advocate them as something more than mere noise in need of reduction. In other words, an implicit contention seems to be that novelty effects are only relevant as a secondary variable, insofar as they can be isolated and extracted. And this, I will argue, is potentially problematic when we consider the fact that the novelty of social robots is essentially *why* we are doing SHRI research in the first place.

In the next section, I will present a tentative account of novelty as an 'original feature of experience'; what I will call experiential novelty. Of this account, the main claim is that given the nature of the experience of novelty as basically a lack of 'pre-established meaning', any cognitive or behavioural effects stemming from this experience should thus be understood as expressions of this lack. Thus, novelty effects are posited to be symptoms, even generators, of learning in a way that makes visible the poverty of common conceptions of them as simply temporary 'noise'.

In section 3, I will argue the explanatory fruitfulness of this account for different research lines within SHRI, and further point to some areas where the account may be employed as an additional source of reference in generating future research.

Lastly, in section 4, I will address some limitations of my arguments as well as some directions for future research.

II. Novelty as an 'Original Feature of Experience'

In this section, I will set out by offering a brief overview of how novelty has been investigated and conceptualized within various research fields so far. After this, I turn to introduce a tentative account of what will be called 'experiential novelty'.

A. *Examples of novelty-treatments in research*

Within most literature on behavioural research methods, definitions of the concept of the Novelty Effect denote the underlying phenomena as potential threats to the validity of a given study on grounds of these effects confounding the generated data. For instance, [20] describe the phenomenon as "[…] increased motivation, interest, or participation on the part of study participants merely because they are undertaking a different or novel task, thereby affecting their responses in a way that is not related to the independent variable". Similar examples can be found in [21, 22, 23]. As is evident in the extracted description, novelty effects are often depicted as a type of behavioral disturbances or noise, distinct from that which is being investigated; a depiction that finds resonance in the previously presented examples from SHRI research.

The notion of novelty effects as an increase in interest is also found within Consumer Research. But contrary to methodological evaluations, the effects are within this domain viewed as something very central to the 'independent variable' by being the expression of a "[…] demand for novelty as the portion of demand for a product or service that is contingent on its perceived newness and distance from other products in product characteristic space. […] demand for novelty is particularly intense in so-called 'hit-driven' industries; fine arts, games, movies, music, television, theme parks, toys, etc." [24].

Nearly 70 years ago, Berlyne [25] presented one of the first treatments of the relation between curiosity and novelty, arguing novelty as a main motivational factor in behavior. This claim is today highly corroborated, and some have recently proposed that the 'need for novelty' may be a fourth basic and innate psychological need in addition to Competence, Relatedness, and Autonomy, as originally posited in the Self-Determination Theory [26].

However, novelty may not always lead to increased interest. Within psychology, the distinction between two main categories of behavioral responses, exploration (sometimes called 'approach') and avoidance, is found at various levels of research [27, 28, 25]. Very simplified, the exploration of novel stimuli and environments (a positive novelty bias or novelty-preference, or in a strong sense, neophilia) is linked to a motivation for seeking out desirable, sometimes 'rewarding', outcomes, and the avoidance of novel stimuli and environments (a negative novelty bias or familiarity-preference, or, in a strong sense, neophobia) is linked to a motivation for avoiding aversive, sometimes 'harmful', outcomes [28, 29].

Several theories of personality and temperamental dispositions also operate, in some form or other, with a notion of novelty: most notably, Cloninger [30, 31] operates specifically with a personality-dimension of Novelty Seeking as part of the Psychobiological Model of Personality. In this model, the trait of Novelty Seeking 'captures the extent to which individuals respond to novel stimulation or situations with exploratory activity' [28]. Similarly, in the more widely known Five-Factor Model (FFM), the trait of Openness to Experience points, amongst other things, to the degree to which an individual is curious and open to novel experiences [32].

Berlyne [25] further noted that responses to novelty are difficult to investigate through the complicated expressions of human behaviour, and therefore often more easily studied in lower animals. This still seems to be the case when considering that much of the accumulated research on human novelty processing and responses since then draws heavily on comparative psychology in general [33], and on research on rats in particular [34]. Consequently, it does not seem unreasonable to speculate whether the conception of novelty effects as 'simply' expressions of increased avoidance or exploration might be an unfortunate derivative from a history of investigating novelty responses through research paradigms that rely on heavily reduced behavioural complexity. I use the term 'unfortunate' here, because, as I voiced earlier, such a conception may potentially misdirect attention away from the vast multitude of cognitive and behavioral expressions that novelty effects take on in 'real life' – a concern that has recently been raised by behavioural ecologists as well [29]. I shall return to this at the end of this section.

B. *Experiential novelty*

While an exhaustive theory of novelty and novelty effects is outside the scope of this paper, a general comprehension of the basic mechanisms involved remains essential.

First, an important distinction needs to be made: there are two fundamental senses in which we can say that something (henceforth X) is novel. Intuitively stated, we can mean either (1) that X is novel *in itself*, i.e. that we are saying something about X itself, or (2) that it is novel *to someone*, i.e. that we are saying something about the experience of X. For reasons of clarity, I will henceforth refer to the former as 'ontological newness', and to the latter as 'experiential novelty'. It is further important to note that these fundamental kinds do not necessarily imply each other: something can be novel to someone, *regardless* of whether or not it is new in itself. Conversely, something can be new in itself, *regardless* of whether or not someone experiences it as novel. In this paper, I have so far referred to novelty in the experiential sense and will continue to do so throughout. I base this choice upon the simple claim that in order for anything to elicit any cognitive and behavioral effects on grounds of novelty, it is only necessary that it is *experienced* as novelty. Thus, the distinction is helpful because it efficiently brings to light the nature of novelty effects as originating in a particular relation between X and the 'experiencer' [33, 35].

> The simplest translation of our commonsense idea of novelty into a more precise notion is that the novelty of an event is assessed by examining a memory store of past observations, where a memory system might require more than one experience of an event to form a lasting memory. An observation is completely novel, to use Berlyne's term, if a representation of it is not found in memory. If memory fades with time, this process assesses short-term or long-term novelty depending to [sic] the fading rate. This of course ignores many aspects both of novelty and of memory, and it may not be feasible from a computational perspective. [36]

This is indeed a very simple description, and as the authors note, it lacks a thorough clarification of the workings of memory for it to be considered a functional explanation. However, for the aims for this paper it suffices, and the basic mechanism of novelty as arising from a lack of match between an experience and memory is consistently found throughout most of the literature on novelty detection and assessment (see [36, 37, 35, 38, 33] for detailed literature reviews). Assuming an intuitive notion of experience and memory, we might then say that *novelty arises as a feature of experience, when we encounter something that we cannot make sense of solely in terms of what we already know.*

This implies novelty as a negation of familiarity, or in other words, that novelty and familiarity reflect two endpoints of a linear continuum [39, 33]. As such, novelty (and familiarity) exist as gradient features; an increase in the degree of familiarity means a decrease in the degree of novelty, and vice versa. Hypothetically, absolute novelty (or 'complete novelty' as in the above quotation) is taken to be something that is completely devoid of any familiar aspects, and conversely, absolute familiarity is taken to be something that is completely devoid of novel aspects. However, as Berlyne also posited, anything experienced as novel must in some way contain an aspect of (or relatable to) something familiar in order for it to enter experience at all [36]. Similarly, it may be suggested that most things experienced as familiar usually contain some aspects of novelty however miniscule. But a further discussion of this touches upon deeper issues of cognition and the process of human conceptual experience that must be saved for another occasion.

From this description we can draw three important insights about novelty. (i) First, novelty is essentially an 'original feature of experience'. (ii) Second, novelty experiences may differ in intensity. (iii) Third, any decrease of novelty involves familiarization, and furthermore, the extent of familiarization needed for something to become not-novel must in some way be related to the intensity of the experience. To elaborate:

(i) If novelty is the negation of familiarity (or from a processing-perspective, a 'deficiency-state' of repetition), then novelty must be the original feature of everything that we experience, which only afterwards, through repetition, gains familiarity [25]. At first, it may seem contra-intuitive to describe novelty (and not familiarity) as the 'default' state of experience, as it is usually the one of the two that requires that *extra* effort of us in our daily lives. But with this description, it hopefully becomes clearer to the reader why we need to dispose of the conception of novelty effects as simply additional noise distorting our 'true' responses: instead, the presence of novelty effects should be understood as indicators of the fact that no 'true' responses have been formed yet. This will be tied to the remaining two insights shortly.

(ii) If an experience of novelty is defined as an inability to make sense of the experience on basis of what we already know, then the intensity of the experience should be correlated with the extent to which our already possessed knowledge proves inadequate. As such, instances of mere surprise may not necessarily lead to novelty as this only entails a violation of expectations that may quickly be remedied by employing additional, already possessed knowledge – an observation supported by the fact that novelty and surprise do not seem to always activate the same neural circuits [36, 35].

The inadequacy of possessed knowledge is hypothesized to rely on the constitution and/or complexity of the novelty-experience: *constitution* points to the degree of novelty/familiarity of the experienced constituent features of X. Berlyne distinguished between absolute (not to confuse with my previous use of the term) and relative novelty, where X is *absolutely* novel if some (or all, I add) features of X have never been experienced before, and X is *relatively* novel if all features are familiar, but constitute a novel whole [36].

Complexity points to the number of features constituting the experience of X. [36] investigate three kinds of novelty-complexity (stimulus, associative, and contextual) usually employed within neuroscience (see [33], for further examples of these), of which they argue that only stimulus and some instances of associative novelty are actual examples of novelty and not mere surprise. Due to space-restrictions, I shall only present the actual notions of novelty:

Stimulus novelty refers to novelty constituted by an experience of relatively simple stimuli, where equally simple, one-to-one representations are lacking. This is the kind of novelty that is usually employed in neuroscientific investigations of repetition effects and habituation.

Associative novelty refers to experiences where larger (usually familiar) bundles of stimuli in combination constitute novelty. [36] argue that true associative novelty only arises if the combination result in the experience of the resulting whole (or *gestalt*) as novel (similar to Berlyne's relative novelty). As such, the distinction implies that if the experience can be adequately 'made sense of' by knowledge of the familiar features, then the experience is not coloured by novelty. But if the combination of the features results in the experience of a gestalt that cannot be adequately made sense of simply by reference to the constituting features, then experiential novelty ensues.

In addition to the constitution and complexity, I further hypothesize that the intensity of novelty-experiences is moderated by the circumstances of the experiencer's practical engagement with X. In conjunction, interrelated factors such as the context [40, 38], intentions and goals [38, 35], perceived affordances in the surrounding environment [41], and the like, may all influence the extent to which the experiencer is able to employ already possessed knowledge in making sense of an experience. E.g., for me, having never been in close proximity with an MRI-scanner before in my life, the intensity of the experience as novel would be far greater if I was suddenly required to use one, than if I was only supposed to stand next to one as a relative to the patient inside.

(iii) If a decrease of novelty is inherently an increase in familiarity, then the process of novelty-reduction is essentially a process of familiarization; basically, familiarization is a process of learning qua the formation of new memory-contents, which in turn reduces the experience of novelty [39, 33, 42]. Thus, the degree (and contents) of the familiarization needed before X is no longer experienced as novel must in some way depend upon the circumstances generating the experience of novelty, i.e., the circumstances of intensity just described. This characterization accords well with the Novelty/Encoding Hypothesis [1] which suggests that '[o]n-line information, including that provided by the sensory systems, is transmitted for higher level encoding on the basis of its novelty: novel information receives preferential treatment over familiar information. Completely redundant

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[1] This hypothesis has been criticized for employing a notion of novelty processing and familiarity processing as relying on two distinct neural circuits [33, 35]. To this I will only observe that although the arguments seem plausible, the proposed conclusion that 'novelty processing' might be a redundant, conceptual construct seems to have overlooked the primacy of novelty: strictly speaking, it should be the construct of 'familiarity processing' that is redundant in the sense of it being simply an inhibition of activity in (novelty) processing.

information is screened out from further processing and is not stored' [42].

Taken together, these three insights carve out a picture of novelty effects far removed from that of distorting noise: if experiential novelty is understood as arising from a failure to make sense of an experience, then novelty effects can further be understood as any cognitive and behavioural responses activated or generated by the experience of novelty. In this sense, *if activated by an experience of novelty*, the process of learning itself becomes a kind of novelty effect. Not only does this suggest that novelty effects are significant since they indicate that 'true' responses have not been formed yet, it also points to the observation that expressions of novelty effects may not necessarily fade away, as commonly assumed. In fact, they may instead become consolidated as precisely the 'true' responses that was initially missing.

1) Exploration and avoidance

Earlier I have argued that a reduction of novelty effects to simple expressions of either exploration or avoidance does not do justice to the vast variety of ways in which responses to novelty may come to expression through human behaviour. However, in terms of novelty effects as a source of learning, the categories of exploration and avoidance may still serve as useful ways of interpreting the 'strategies' underlying the multitude of behavioral novelty-expressions observed.

When the formation of novel memory-contents is required in order for us to make sense of something, it means that we have to seek out other sources of information than what our memory-contents provide. As such, while our memory-contents may to a certain extent guide it, our engagement with our surroundings becomes the means to this search. A 'symptom' of being in this search is that predictions are only made tentatively, i.e., not confidently or automatically relied upon. This explains why experiencing novelty is often described as being in a state of uncertainty [38]. Accordingly, I suggest that whatever kind of particular 'search-strategy' we may employ (consciously and subconsciously), it might be helpful to understand it as relating to a more general intention of either exploring X or of avoiding X. Common for strategies relating to the intention of exploring X is the attempt to gain knowledge of X, for instance through asking questions, taking it apart, imitating other's engaging with it, etc. Common for strategies relating to the intention of avoiding X is the attempt to reduce the need for knowledge about X, for instance through removing oneself from X, ignoring X, asking someone else to handle X, etc.

Whenever a strategy proves successful, it seems reasonable to assume that it will be consolidated into memory and once again employed next time X is encountered. What is important in this description is the illustration of how initially novelty-generated assumptions and behavioural expressions may turn into lasting ways of thinking and responding. Albeit very intuitive, the description is compatible with the general notions of exploration and avoidance as described earlier, as well as with several theories of learning, including the theory of action-based cognitive dissonance-reduction [43], and theories on the formation of scripts and schemata [44].

C. Summary

In sum, the reframing of novelty effects that I propose in this paper is centered upon these four claims: 1. Novelty is an 'original feature of experience', influencing in various degrees and intensity how human beings engage with their surroundings; 2. Cognitive and behavioural novelty effects arise as an experiencer engages with X in a way that renders her already possessed knowledge inadequate in making sense of X; 3. Because the experience of novelty is (at least at a certain minimum degree of intensity) followed by the generation of additional knowledge, novelty effects should be understood as both the means to and expressions of this generation; and 4. In instances of research where engagement with the investigated subject may give rise to novelty experiences, the presence of novelty effects should be investigated not as an unwanted source of noise, but as a valuable source of information on how the subject is made sense of.

III. Novelty effects within social HRI

I will now turn to discuss the usefulness of this reframing within research of SHRI. I will set out by arguing why social robots may be experienced as novelty today. After this, I discuss examples of SHRI areas and topics in relation to experiential novelty, in order to illustrate how a reframing of novelty effects may be of use for analysis and explanation in SHRI.

It is important to note that given space restrictions, this discussion is aimed at arguing the promise of a reframing of novelty effects within SHRI in general. This means that I focus on common topics and observations, rather than specific sets of studies or disciplines, in order to illustrate the broad scope of the account's relevance. However, given that SHRI is undertaken from different disciplinary perspectives, a more differentiated discussion of the respective relevance of the account suggested here is an important next task.

A. Social robots

By a *social* robot, I mean a "[…] programmable artifact that is designed to act within the physical and symbolic spaces of human social action in accordance with social norms; in addition, many such devices are intentionally designed to exhibit forms of behavior that invite--afford or even elicit--human social engagement" [45]. I do not focus on functionality in itself, but instead on the *experience* of the robot as capable of reciprocal social interaction (similar to the perspective proposed by Coeckelbergh [46]).

However, it is important to keep in mind that although I argue the experiential novelty of social robots with reference to a strong notion of sociality, I do not mean to imply that robots experienced as less socially capable may not potentially be experienced as novelty; only that the more socially capable, the more intense the experience of novelty is expected to be. In addition, I also do not mean to imply that experienced sociality is the only feature of a robot that could cause it to be experienced as novelty. In theory, the 'trigger' could be any feature that the robot is experienced to have, depending on the knowledge and engagement of the experiencer. In the following, I am simply offering an account of why social robots *per se* may be experienced as novelty in a way that could plausibly be expected to hold for most people today.

B. The experiential novelty of social robots

The introductory quotation [1] illustrates effectively the unclarity with which novelty is usually attributed. With their use of the term 'ontological' it seems that the researchers intend to suggest that social robots as a category may be an ontologically new category of beings, which would mean a category of beings that is new solely by the fact that it has not existed before. However, what they go on to describe are clearly characteristics of experiential novelty. For the purposes of this paper, the question of whether or not social robots may be ontologically new holds little relevance. Yet, it should be noted that to confuse the nature of the two may lead to equally confused methods of identification. As experiential novelty essentially says something about the *experience* of something (e.g., a social robot), identifying such an attribute solely by investigating the features of this something *in itself,* is highly problematic – and equally so in the reversed scenario. This is illustrated when [47] argue that the truth of the proposition that social robots represent a new ontological category depends upon whether or not people are actually experiencing these robots as both animate and inanimate or only pretending to do so. However, if one aims for justification that social robots are ontologically new, then it becomes irrelevant what people believe. If, on the other hand, one aims for justification that social robots are experientially new, then the distinction becomes relevant in a different way than intended by [47]. In this case the distinction points to different degrees of intensity in the potential experience of novelty: if they are believed to be simultaneously animate and inanimate, then it would most likely imply a need for revising the basic structures of our assumptions about animacy and inanimacy, which probably gives rise to a very intense experience of novelty. If they are only treated 'as if' they are simultaneously animate and inanimate, it may imply a less severe need for simply updating these assumptions with additional information on possibilities for engagement, which would probably give rise to a less intense experience of novelty – but chances are that novelty would arise nonetheless, as most would probably still have no experience with anything inanimate affording this kind of *interactive* pretense.

If understood as referring to experiential novelty, the introductory quotation points directly to that characteristic of social robots which I argue is highly likely to give rise to the experience of novelty; namely, the unique combination of features (of animacy and inanimacy) that are usually found to be mutually exclusive. As indicated, I will further argue that it is the experienced *sociality* of this combination that renders our knowledge of the features individually inadequate, because the combination may be experienced as a 'novel, complementary form of social interaction' [48]. As a result, we are unable to make sense of the experience through prior knowledge, and likewise unable to rely with certainty on any predictions we might make in our engagement with the resulting gestalt.

Another way of arguing social robots as experiential novelty, is found in [3], where Swan reflects the following:

> In some ways technology is still man's tool, but in other ways perhaps technology can no longer be subsumed under this category since it is increasingly standing on its own as an existence in the world. Thus, there is an emerging triumvirate view of reality as nature, man, and technology which is forcing man to adapt and redefine himself in new ways. [3]

In referring to the generalized concept of 'man', Swan touches upon a dimension of experiential novelty that pertains to a larger experiential system transcending the boundaries of the individual. In section 2, I focused on individual experiences of novelty. However, the arguments from this account may also be relevant to reflect upon in relation to this larger frame of perspective: with the increasing integration of embodied robots as social agents within societal and cultural domains, collectively shared and accumulated knowledge of what it means to be an agent within these domains may equally become challenged to an extent where this knowledge will become inadequate. In this way, social robots may also constitute novelty in their relation to larger system of collectively shared and enacted experience. If so, then it becomes important to consider the ways in which novelty effects of this kind may find expression. I shall return to this again at the end of this section. In the following, I will continue to place the discussion at the level of the individual.

C. Anthropomorphization as a novelty effect

A relatively straightforward example of novelty effects at large within SHRI research, is the phenomenon of anthropomorphization. In [49], the authors propose that one of the motivations for anthropomorphizing lies in the human need for effectance, in that the attribution of 'human characteristics and motivations to nonhuman agents increases the ability to make sense of an agent's actions, reduces the uncertainty associated with an agent, and increases confidence in predictions of this agent in the future' [49]. Therefore, observed anthropomorphizing could be a symptom of an experience of novelty. If so, then anthropomorphization may in such instances not necessarily say anything about the intrinsic nature of social robots, but only about the fact that in the observed situations, the participant's engagement with the robot affords this kind of response.

This further becomes interesting, when considering that [49] further propose that a second motivational factor is the human need for sociality. From a perspective of novelty effects as noise, it would seem sensible to hypothesize that if the anthropomorphization was brought on by a need for effectance, (i.e., as a novelty effect) then this should disappear as the individual gradually becomes familiarized with the robot. Consequently, if it does not disappear over time, this suggests that the anthropomorphization is employed not in the need of effectance, but in the need of sociality. Such a conclusion could then further imply that social robots are anthropomorphized either because they intrinsically pose a fitting opportunity to satisfy a need for sociality, or because they are experienced as actually possessing characteristics and motivations initially thought to be only anthropomorphized onto the robot. However, remembering that expressions initially generated as novelty effects may not necessarily fade away, such conclusions would also be premature as the continued anthropomorphization might also simply rely on the fact that it has proved successful in providing the experiencer with a way to make sense of and engage with the robot. As such, even observed increases in anthropomorphization [5] could be explained by the participants becoming comfortable with the employed strategy and its entailing narrative of the robot as humanlike, having thus reduced the intensity of experienced novelty.

D. *Novelty effects as failed expectations*

Several have voiced the importance of designing the physical appearance of social robots in a way that reflects their functional capabilities, so as to provide people with more sustainable 'points of entry' into the interaction [6, 50]. The reasoning behind such proposals becomes further fleshed out through the presented account of novelty-responses: the better the physical appearance is at providing people with accurate cues on what kind of engagement the robot may hold, the more likely it is both (1) that the intensity of experienced novelty is proportionally aligned with the needed familiarization, and (2) that the 'strategies' chosen in this familiarization proves successful. The importance of this proportional alignment resides in the fact that novelty is essentially about being aware that there is something one does not know. However, the extent of this knowledge-lack may not always be clear, which means that something may initially seem to us as more (or less) novel than it actually turns out to be. Thus, the alignment may help to keep unwarranted emotional appraisals in check. As noted in the introduction, [14] reflect that the observed radical decrease of participant interactions with the robot may have been due to the fact that the participants initially formed higher expectations of the interaction than could be sustained by the robot, thus leading to disappointment. In this sense, the initial 'strategies' chosen did not prove successful, which may have, as [5] suggests, led to negative evaluations of the following interaction in a way that could have been avoided. These reflections compliment well the argument by [17], that the adoption of robotic technology should be understood from an expectancy-confirmation model.

E. *Novelty effects as determined by engagement*

In relation to this, the experiential novelty-account also points to the role of the way in which participants engage with the studied robots. Some studies on attitudes towards social robots measure attitudes in settings where there is no direct interaction between the participants and the studied robots [18, 50]. As such, these studies measure attitudes that are based on the participants' immediate conceptions of what a social robot may be; conceptions that remain relatively unchallenged and static throughout the experiment. However, it seems plausible to assume that if social robots are experienced as novelty qua their experienced sociality, then such studies fail to recognize the challenge in sense-making that may occur through actual, dynamic engagement with the robot [6]. The conclusion of [50] that they expect the results of their study 'to have an impact on long-lasting psychological and philosophical debates regarding human-machine interactions' thus seems to be a bit of a stretch.

Overall, the account of novelty experiences as influenced by the circumstances of the human-robot engagement may provide a further frame of reference from which to investigate these circumstances. The role of robot-embodiment is one such circumstance. Studies have found that people seem to react differently to on-screen or virtual robotic simulations vs. physically present, embodied robots [51, 7]. A further way of investigating these differences could be through the acknowledgement that different levels of embodiment equally pose different affordances of 'sense-making' – and that novelty responses, as sense-making activities, may equally come to expression differently at these various levels. A physically present robot may presumably be experienced as possessing a greater extent of autonomy [7] and consequently, as offering a greater range of possibilities of engagement. If so, then it seems plausible that a greater extent of autonomy will be experienced as requiring a more complex understanding of the agent, thus leading to more intense novelty-experiences as well as more tangible novelty effects, as similarly proposed by [5].

F. *Implications of the consolidation of novelty effects*

The suggestion of novelty effects becoming consolidated into existing knowledge could equally hold analytical utility for SHRI research. For instance, as noted in relation to anthropomorphism, observations of certain ways of engaging with social robots may not necessarily indicate an intrinsic nature of these robots, but instead only successfully employed strategies of sense-making. The difference is essentially a difference between observations of possibilities and of necessities, between *can* and *should*; within ethical debates, this difference is highly relevant [52]. Even though studies have found that people seem to treat robots in a way that suggests that these robots *can* be made sense of moral recipients, the question is whether or not such sense-making *should* be encouraged. A discussion of the different vocations within this area remains outside the scope of this paper. Yet, I do offer the brief suggestion that analyzing the different strategies of sense-making employed in social human-robot engagements from a framework of experiential novelty may reveal a more detailed account of the workings of these strategies. In this way, we might be able to distinguish between strategies employed from necessity and strategies employed from possibility. These insights could then be used in our efforts to strengthen and encourage some strategies while redirecting others, in the attempt to ultimately guide what kinds of representations that become consolidated into long-term knowledge – this suggestion resembles the proposal by [15] to 'identify the social behavior parameters that are influenced by habituation effects in order to create better social behavior models for robots that adapt their social rules over time'. However, it differs in that I propose the employment of such knowledge in equally creating better social behaviour models for *humans* engaging with these robots.

G. *Collective novelty effects within social HRI*

This last proposal touches once again on the role of collective novelty effects. As social robots become increasingly integrated into society and socially shared settings in general, they are 'made sense of' in a way that transcends the single experience of the individual. This, if anything, underlines the centrality of novelty within SHRI research: if attempts to generate meaning on account of novelty are understood as expressions of novelty effects, then collective attempts to generate meaning on account of novelty could equally be understood as such expressions. If so, the field of SHRI itself can be understood as a generating expressions of novelty effects. Thus, in asking what kinds of robot-representations should be encouraged and which ones should not, we, as researchers, need to acknowledge our own role in shaping the answers that we may find. In [13], the researchers voice this concern in relation to how current framings of 'assistive robots' reinforces implicit tenets of ageism, and [53] notes in relation to child-care robots:

> Ethical appraisals of robot-assisted childcare, however, are hindered by a lack of empirical evidence about long-term impact. Consequently, the topic area becomes a depository of hopes, fears and even fantasies. These find expression in the public domain in the ways that news media, technology and parenting blogs, as well as in companies' promotional material, disseminate technological innovations. [53]

In turn, these collective novelty effects modulate experiences and expressions of novelty at the level of the individual [4]. Therefore, it becomes problematic when researchers propose conclusions like: "A robot that appears female and humanlike is more likely to be perceived positively […]. Conversely, a robot intended to create fear for entertainment purposes would likely benefit from having male, machinelike features" [18]. Although the observations may be valid, the conclusion drawn is an example of confusing *can*-observations with *should*-observations: the fact that female and human-like robots are observed to be more positively perceived than male and machine-like robots, does not necessarily mean that such perceptions *should* be encouraged. It may only be that what is observed are expressions of a regression to historically reinforced gender-stereotypes in situations where no other knowledge seems adequate; a regression that is now viewed as problematic even in human-human interaction.

IV. Conclusion

In the last section, I have proposed that social robots, experienced as reciprocally social interaction partners, may be experienced as novelty because their social capabilities render conventional knowledge of animacy and inanimacy inadequate in making sense of our engagement with them. As such, the experiential novelty of social robots lies not only on the fact that most people have never interacted with a such an agent before, but also on the fact that most people will likely experience a gap in their already possessed knowledge when they *do* interact dynamically with them. A gap that will need to be filled in one way or another. Therefore, I have argued that the phenomenon of cognitive and behavioural novelty effects lie at the very heart of contemporary SHRI research, and should be addressed not as noise in need of reduction, but as valuable sources of information central to what we seek to investigate. Although not exhaustive, I have provided some examples of this through a discussion of potential novelty effects within various research lines of SHRI research. Most notably, I offer an explanation of anthropomorphism as a novelty effect, and demonstrate how the notion of novelty effects as expressions of sense-making can be employed in offering further explanatory utility in various findings. Thus, the tentative account of experiential novelty has promise as a framework for retrospective analysis, when employed in re-interpretations of already attained findings.

In addition, the account could also hold utility as a frame-of-refence from which to inspire new research-questions, such as: (1) What role does the experimental narration play in providing the participant with cues on how to make sense of the robot? (2) To what extent is the engagement observed influenced by exploratory or avoidance-intended 'search strategies'? (3) To what extent can an engagement be challenged before such search-strategies take over? (4) Is the sense-making underlying our observations desirable in the longer run? (5) To whom does it have to make sense, and why?

V. Limitations and directions for further research

As I contend that novelty remains (for now) an inherent part of SHRI research in general, the areas of research that I have focused on in this paper only scratches the surface of where the account of experiential novelty may be relevant. Most notably, research on the correlation between temperament/personality dispositions and attitudes towards social robots will be an interesting field to explore in terms of novelty effects. Especially, as several studies report not to have found significant correlations between the trait of Openness to Experience and robot-attitudes [54]. However, it remains unclear to what extent measures of this trait actually imply novelty in the sense of unfamiliarity, and not simply surprise, salience or rarity in general. In addition, the presented account of experiential novelty offers a far wider notion of novelty responses than common conceptions, and hence it remains unclear to what extent other traits, such as Extraversion which has been found to play a more significant role in human-robot interaction, relate to this notion. This will have to be addressed in further research.

A possible critique of the presented account could be that, ironically, there does not seem to be anything radically novel in the mechanisms described. In other words, it builds largely on already established areas of research, and generates, likewise, insights that other frameworks may lead to as well. So why bother? I believe that the utility of the account lies in exactly this: building bridges between various findings and theories in a way that make the pieces 'fit together' more comprehensively, revealing details and nuances that may potentially be otherwise overlooked. In the end, I propose that the fact that the experience of novelty is an inherent pre-condition of experiential existence makes any account of the outcomes of such existence not addressing the role of novelty incomplete.

However, the account remains tentative, and much more research is needed in order for it to truly hold explanatory utility - and not only within the domain of SHRI research. Novelty experiences like the kind that social robots may elicit do not come around every day. As such, I believe that investigating the mechanisms of novelty effects through social human-robot interaction equally provide a unique opportunity to study these mechanisms in more detail.

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